Part 6: Preparing for the Landing on Planet 1

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Before landing, we study planet 1 from orbit.

- **Low orbit.** A Hohmann descent brings the spacecraft from 18,992 km to a circular orbit 360 km above the surface. We choose that altitude by measuring the air drag directly: we flew test orbits at 31 altitudes and recorded how fast each one decays. The decay falls off with a scale height of 9.26 km.
- **Composition.** A χ^2 search of a 10-million-point spectrum, calibrated against pure noise, detects CH_4, CO and H_2O. All their lines share a Doppler shift of +0.781 km/s. O_2 is marginal in the spectrum.
- **Atmosphere model.** The model is adiabatic up to 33.2 km and isothermal above. We test it against air densities measured from the orbital decay. These require a mean molecular weight of $\mu = 23.63 \pm 0.02$, which agrees with the composition that includes O_2 ($\mu = 23.69$) and rules out the one without it ($\mu = 20.83$, 39× too dense). The same drag data favour uniform gravity in the model.
- **Spin.** Drag on a retrograde test orbit measures the planet's spin, giving a rotation period of 18.41 h (catalogue: 18.51 h).
- **Landing site.** We derive the coordinates of a point on the spinning surface at any time and scout one orbit of pictures. We choose an equatorial site where the star stands 69° high at the next pass overhead.

I. INTRODUCTION

Landing needs three things we do not yet have.

- A starting orbit close to the planet (Section II).
- A model of the atmosphere, which will brake the lander. It needs the composition, from a spectrum (Section III), and a physical model of how density falls with height (Section IV).
- A landing site, with its coordinates on a spinning planet (Section V).

Throughout, we command the spacecraft through `ast2000tools`' `LandingSequence` [1]. It works in SI units relative to the planet's centre and does not track fuel, so we keep our own fuel account with the rocket equation. We start from the Part 5 orbit with 4,284 kg of fuel. Solar system seed: 42716.

II. ENTERING A LOW ORBIT

A. Method

For the descent from r_1 to r_2 we set the velocity at r_1 to the apoapsis speed of the transfer ellipse,

$$v_{\rm apo} = \sqrt{GM\left(\frac{2}{r_1} - \frac{1}{a}\right)}, \qquad a = \frac{r_1 + r_2}{2}, \tag{1}$$

fall for $\pi\sqrt{a^3/GM}$, and circularise with $v = \sqrt{GM/r_2}$.

How low is safe? Drag removes orbital energy. For a circular orbit in air that co-rotates with the planet (Part 7), the drag force $F = \frac{1}{2}\rho C_d A v_{\rm rel}^2$ with $C_d = 1$ shrinks the orbit by

$$\Delta a_{\rm orbit} = 2\pi\rho\,\frac{A}{m}\,a^2\left(\frac{v_{\rm rel}}{v}\right)^2 \tag{2}$$

per orbit. We therefore probe the atmosphere with the spacecraft itself. We enter circular test orbits at 31 altitudes between 150 and 2000 km and watch each for 10 orbits. These are test runs of the landing simulator; the flight restarts afterwards.

B. Results

Figure 2 shows the measurement.

- At 250 km and below, drag wins and the API ends the run.
- From 250 to 320 km the decay falls exponentially, with a scale height of $H = 9.26$ km.
- Higher up, only the integrator's numerical drift of 0.015 m per orbit remains.

The decay drops to 0.1 m per orbit at 328 km. We put the low orbit 3 scale heights higher, at 360 km, where drag is twenty times weaker.

The descent costs 871.2 and 1224.4 m/s, or 2049 kg of fuel, leaving 2,235 kg (Fig. 1). Over 20 orbits in the new orbit, the semi-major axis changes by only 0.06 m, which is the integrator's drift, and $e < 1.3 \times 10^{-9}$ (Fig. 3).

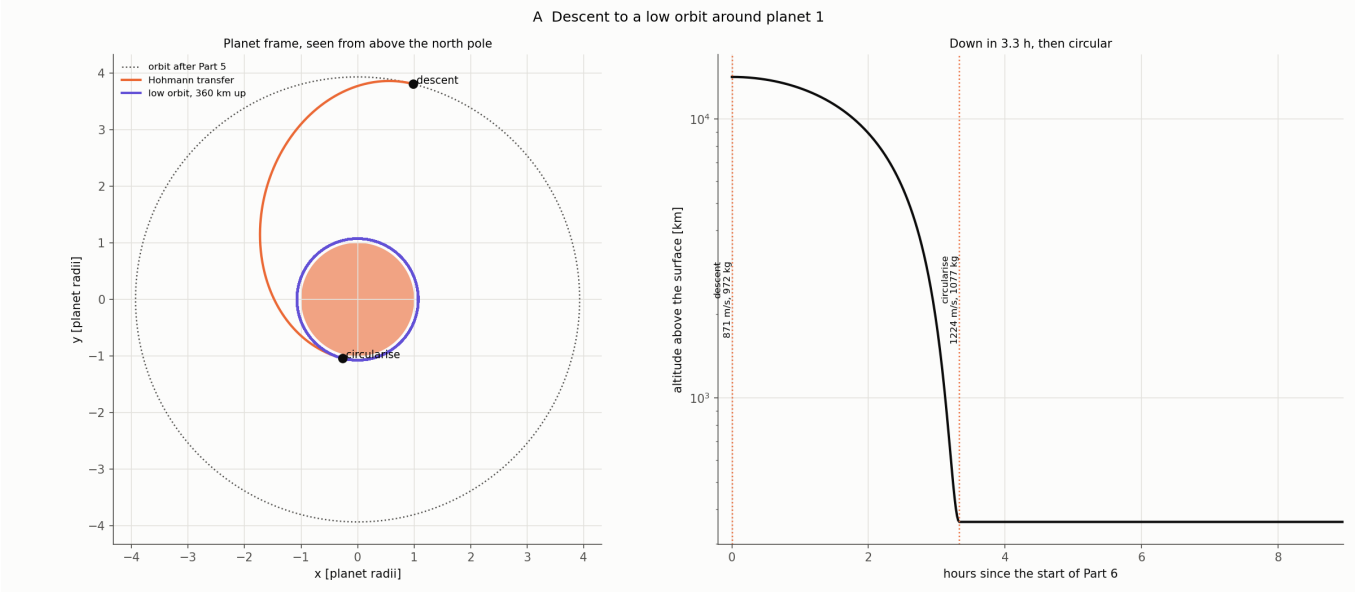


FIG. 1. The Hohmann descent from the Part 5 orbit to the low orbit, in the planet frame and as altitude against time.

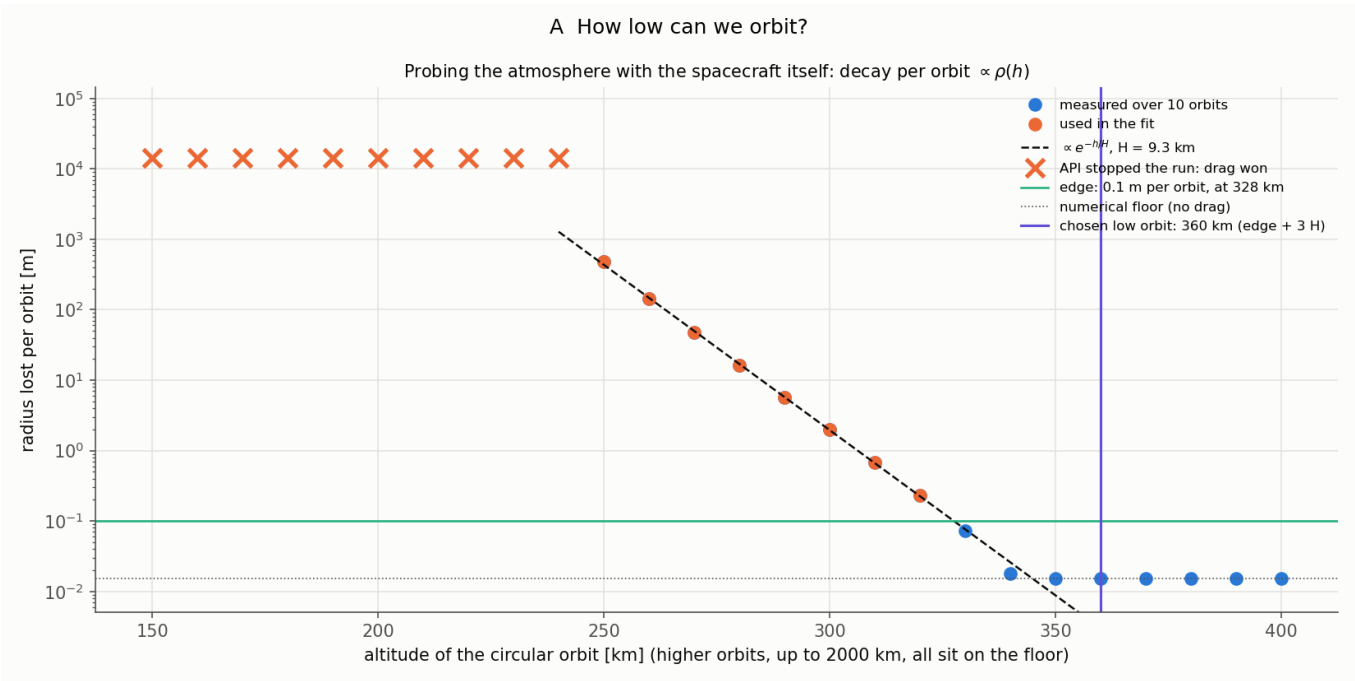


FIG. 2. Drag probe. Radius lost per orbit against altitude, the exponential fit, and the chosen low orbit.

III. SPECTRAL ANALYSIS OF THE ATMOSPHERE

A. Theory

The spacecraft moves at most $v_{\max} = 10\,\mathrm{km/s}$ relative to the atmosphere, so a line at λ_0 can be shifted by at most

$$\Delta\lambda_{\max} = \lambda_0 \frac{v_{\max}}{c}, \tag{3}$$

which is 0.0211 nm at 632 nm and 0.0957 nm at 2870 nm. Each molecule's velocity along the line of sight is Gaussian with $\sigma_v = \sqrt{kT/m}$ (Part 1). Since $\Delta\lambda/\lambda_0 = v/c$, the line profile is a Gaussian of width

$$\sigma = \frac{\lambda_0}{c}\sqrt{\frac{kT}{m}}. \tag{4}$$

For O_2 at 632 nm and 150–450 K, that is 0.00041–0.00072 nm: only 2–3 data points, given the grid step of 0.00024 nm.

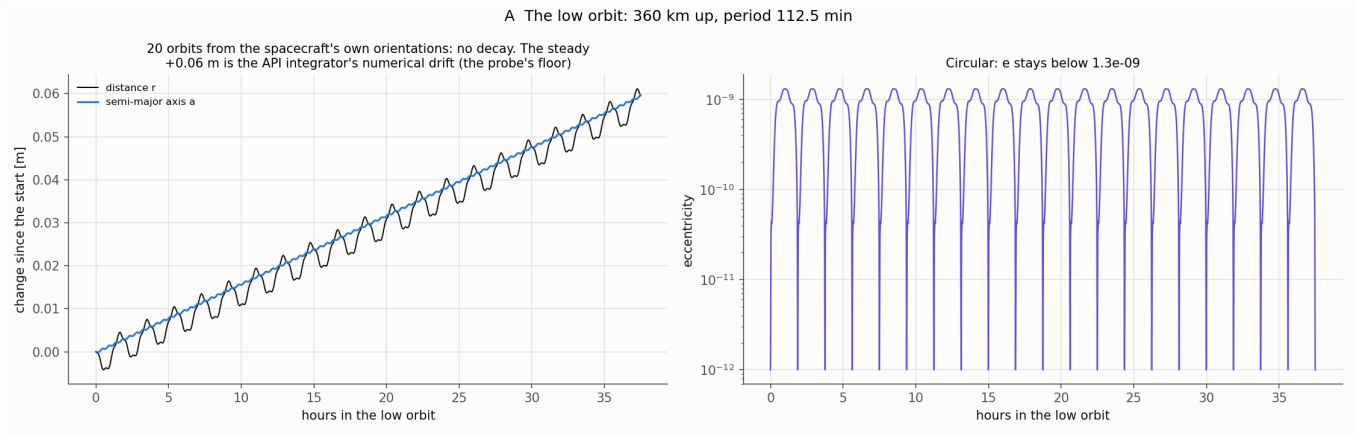


FIG. 3. The low orbit over 20 orbits, from the spacecraft's own orientations.

B. Method

Each line is modelled as

$$F = 1 + (F_{\min} - 1)\, e^{-(\lambda-\lambda_c)^2/2\sigma^2} \tag{5}$$

and fitted by minimising

$$\chi^2 = \sum_i \left(\frac{f_i - F(\lambda_i)}{\sigma_i} \right)^2 . \tag{6}$$

The noise σ_i varies between 0.05 and 0.20 along the spectrum (Fig. 5). Four ideas make the analysis fast and objective.

a. Exact depth. The model is linear in the depth $A = F_{\min} - 1$. For each (λ_c, σ) the best depth is $A^* = S_{fg}/S_{gg}$, with $S_{fg} = \sum w(f-1)g$, $S_{gg} = \sum w g^2$ and $w = 1/\sigma_i^2$. As functions of λ_c both sums are convolutions with a Gaussian kernel. The whole (λ_c, T) grid therefore costs one convolution per temperature. A least-squares fit then refines the minimum and gives uncertainties.

b. Calibrated threshold. A χ^2 fit always finds *some* line. To know what noise alone can do, we run the identical search on 400 line-free stretches of the spectrum near each line. A line counts as detected only if its $\Delta\chi^2 = \chi^2_{\text{no line}} - \chi^2_{\text{best}}$ beats noise with a false-alarm probability below 0.001.

c. Two stages. A blind search over all ± 10 km/s gives the strong lines and their common Doppler shift. A second search looks only within ± 0.24 km/s of that shift. With far fewer places for noise to hide, its threshold is much lower: it corrects for the look-elsewhere effect.

d. Completeness. We inject synthetic lines of known depth into the noise to see how deep a line must be for us to find it.

The procedure is summarised in Fig. 4.

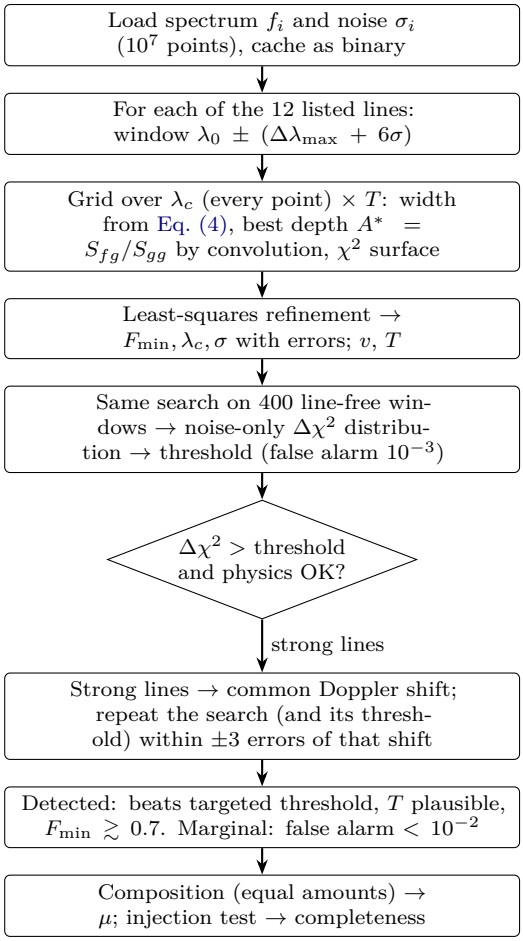


FIG. 4. Flowchart of the spectral analysis (challenge B).

C. Results

The blind search finds 4 lines far above their thresholds, with $\Delta\chi^2$ against threshold:

- CH_4 1660 nm: 57 vs. 22;
- CH_4 2200 nm;
- CO 2340 nm: 101 vs. 19;
- H_2O 940 nm: 23 vs. 19.

All four share a Doppler shift of $+0.781 \pm 0.036$ km/s (Figs. 6 and 8). That agreement alone is strong evidence that they are real: flukes land anywhere in ± 10 km/s. The detected gases are H_2O, CH_4, CO, which gives $\mu = 20.83$.

a. The O_2 question. At 760 nm the fitted line sits at the common shift (+0.79 km/s), with $\Delta\chi^2 = 11$. That is just below the targeted threshold of 12, so it counts as marginal. Its $F_{\min} = 0.66 \pm 0.13$ is below the expected 0.7. It lies where the noise is 1.29 times its median, and noise there can deepen a real line as easily as it can fake one.

The completeness test explains why O_2 is hard to see. All three O_2 lines lie in noisy parts of the spectrum. There, our search finds a line with 90% reliability only if it is deeper than about 0.3 (Fig. 9), deeper than real lines are expected to be. The spectrum therefore cannot exclude O_2, whereas CO_2 and N_2O lines deeper than about 0.13 would have been found. Section IV settles the O_2 question with an independent measurement.

The temperatures from the line widths are poorly constrained (lines only 2–3 points wide). They agree with 150–450 K only within their large errors.

IV. A MODEL OF THE ATMOSPHERE

A. Theory

We combine

- hydrostatic equilibrium, $\mathrm{d}P/\mathrm{d}h = -\rho g$;
- the ideal gas law, $P = \rho k T/\mu m_u$;
- the adiabatic relation $P^{1-\gamma}T^{\gamma} = \text{const}$, with $\gamma = 1.4$, below the height where $T = T_0/2$, and constant T above.

Together these give

$$\frac{\mathrm{d}T}{\mathrm{d}h} = -\frac{\gamma-1}{\gamma}\frac{\mu m_u g}{k} \text{ (adiabatic)}, \quad 0 \text{ (isothermal)}, \tag{7}$$

$$\frac{\mathrm{d}\ln\rho}{\mathrm{d}h} = -\frac{\mu m_u g}{kT} - \frac{1}{T}\frac{\mathrm{d}T}{\mathrm{d}h}. \tag{8}$$

The boundary conditions are $T_0 = 281$ K (Part 3) and $\rho_0 = 9.09\,\mathrm{kg/m^3}$ (`ast2000tools`). With uniform gravity the solution has a closed form: T falls linearly, $\rho = \rho_0(T/T_0)^{1/(\gamma-1)}$, and then ρ falls exponentially with $H = k(T_0/2)/(\mu m_u g_0)$.

B. Method

We integrate $(\ln\rho, T)$ upward with an adaptive Runge–Kutta solver. The solver stops exactly where $T = T_0/2$ (an event) and continues isothermally. `density(h)` and `temperature(h)` interpolate the solution (Fig. 10). Run with uniform gravity, it matches the closed form to 8×10^{-5}.

We then compare the model with the drag probe. Inverting Eq. (2) with the spacecraft's mass and area turns each measured decay into a density (19 heights, 245–335 km). We fit μ to those densities.

C. Results

Figure 12 is the key result.

- With uniform gravity, the drag densities require $\mu = 23.63 \pm 0.02$.
- Only the composition that includes the marginal O_2 agrees ($\mu = 23.69$): the model then matches the measured densities to a median factor of 0.97 over four decades.
- Without O_2 ($\mu = 20.83$) the model is 39 times too dense.

We therefore adopt $O_2 + H_2O + CH_4 + CO$. The two independent measurements agree: a marginal spectral line, and the air density weighed by the spacecraft's orbit.

The same data also decide the gravity question. Letting g fall with height gives a scale height near 300 km of 10.71 km, against the measured 9.26. It needs $\mu = 25.07$, matching no allowed composition, and it fits twice as badly. Gravity is 4.0% weaker at 100 km, a small change. Its effect on density compounds over many scale heights, however (Fig. 13): a factor of 1.24 at 100 km and 6.3 at 300 km. We adopt uniform gravity, as the assignment allows and the data prefer.

The adopted atmosphere (Fig. 11):

- at the surface: 8.97 bar and $9.09\,\mathrm{kg/m^3}$;
- adiabatic up to 33.2 km, cooling at 4.23 K/km;
- isothermal above, at 141 K, with $H = 9.49$ km;
- density $1.41 \times 10^{-3}\,\mathrm{kg/m^3}$ at 100 km and $1.79 \times 10^{-15}\,\mathrm{kg/m^3}$ at our low orbit.

D. After the fact: the API's own atmosphere

Everything above was decided from measurements alone. Afterwards we read the atmosphere that `ast2000tools` uses internally (Fig. 14). These are private attributes, so we kept them out of every decision.

- **Molecule mass:** 23.5 proton masses, or 23.67 u. Our adopted value is 0.06% off, and the drag measurement alone is 0.17% off.
- **Composition:** 23.5 is exactly $(32+18+16+28)/4$, i.e. O_2, H_2O, CH_4 and CO in equal amounts. O_2 is really there.

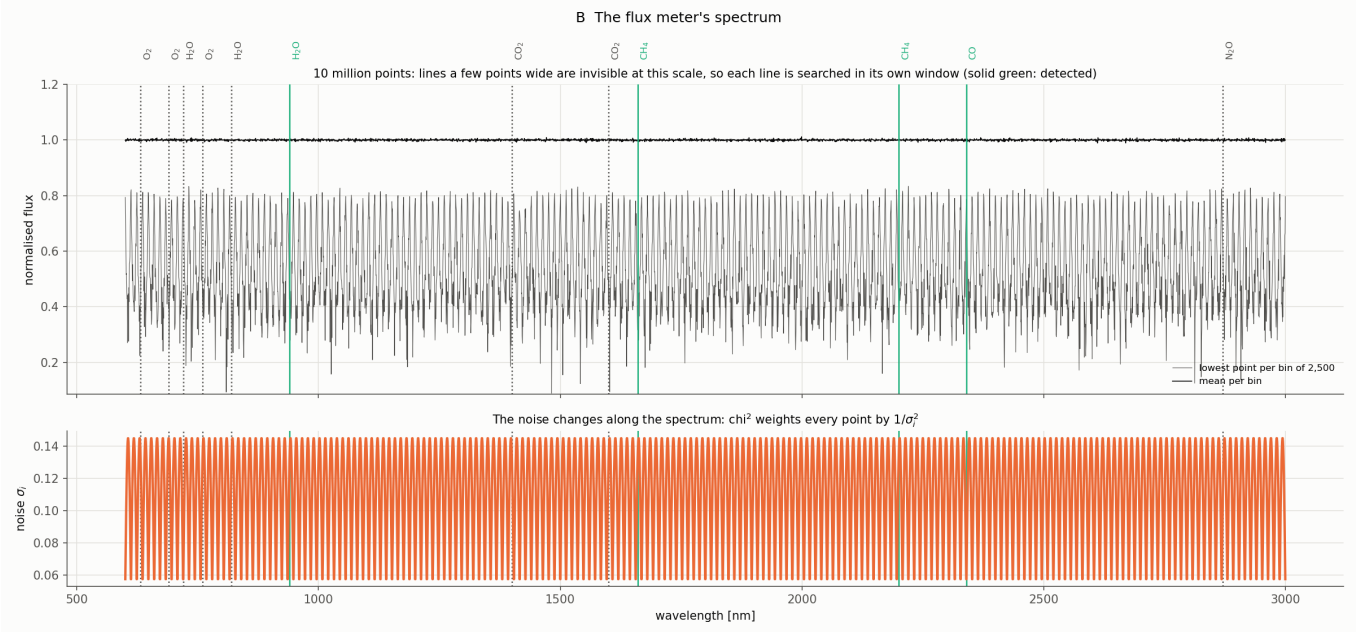


FIG. 5. The whole spectrum (binned) with the listed lines and verdicts, and the noise σ_i along it.

- **Other parameters:** the API uses $T_0 = 281.09\,\mathrm{K}$ (the same black-body formula as Part 3), $C_d = 1$ and uniform gravity.

Our density profile matches the API's to 0.993 at 100 km and 0.980 at 300 km, and the temperature to within 0.09 K. Had we trusted only the spectrum's clear detections, the density at 300 km would have been 44 times too high.

V. SCOUTING FOR LANDING SITES

A. Coordinates on a spinning planet

We use spherical coordinates (r, θ, ϕ) as in Part 4, in the planet-centred landing frame. The planet spins about z with $\omega = 2\pi/P_{\mathrm{rot}}$, so a point fixed on the surface keeps r and θ and

$$\phi(t_0 + \Delta t) = \phi(t_0) + \omega\,\Delta t. \tag{9}$$

The function `new_coordinates` implements this for any number of points and times at once.

Two checks make sure this is right.

- **The frame.** Part 5's last state in the solar-system frame equals the landing sequence's first one to 0.003 m, so the axes are shared.
- **The spin.** At 290 km a retrograde test orbit, which meets the co-rotating air faster, decays 1.49 times faster than a prograde one (Fig. 15). Solving $(v + \omega a)^2/(v - \omega a)^2$ for ω gives $P_{\mathrm{rot}} = 18.41\,\mathrm{h}$, against 18.51 h from `ast2000tools`. That is 0.5% off, and it confirms counterclockwise spin.

B. Scouting

During one low orbit, with the camera pointing straight down, we take 8 pictures (MCAst files `site_k.xml`). For each we record the point below, its planet-fixed longitude, and the star's elevation there, from the planet's position on its orbit (Kepler, Part 2). The spacecraft passes over a given site every

$$\left(\frac{1}{P_{\mathrm{orbit}}} - \frac{1}{P_{\mathrm{rot}}}\right)^{-1} = 2.09\,\mathrm{h} \tag{10}$$

(Fig. 17). The lander will come down near such a pass. For every site we therefore compute the next pass after scouting, and the star's elevation then.

C. Result

4 of the 8 sites are in daylight at their next pass. We choose site 3 on the equator ($\theta = 90°$, planet-fixed $\phi = 312.0°$ at $t = 0$). It is passed overhead at $t = 6.12\,\mathrm{h}$, when the star stands 69° high (Fig. 16), close to local noon. At that time it sits at $\phi = 71.0°$ in the landing frame. The coordinates are saved for Part 7. The pictures can be inspected in MCAst to veto bad terrain; the code takes a manual choice instead if needed.

VI. DISCUSSION AND CONCLUSION

The most valuable tool in this part was one we built ourselves: using the spacecraft's own orbital decay as an instrument. It gave four results.

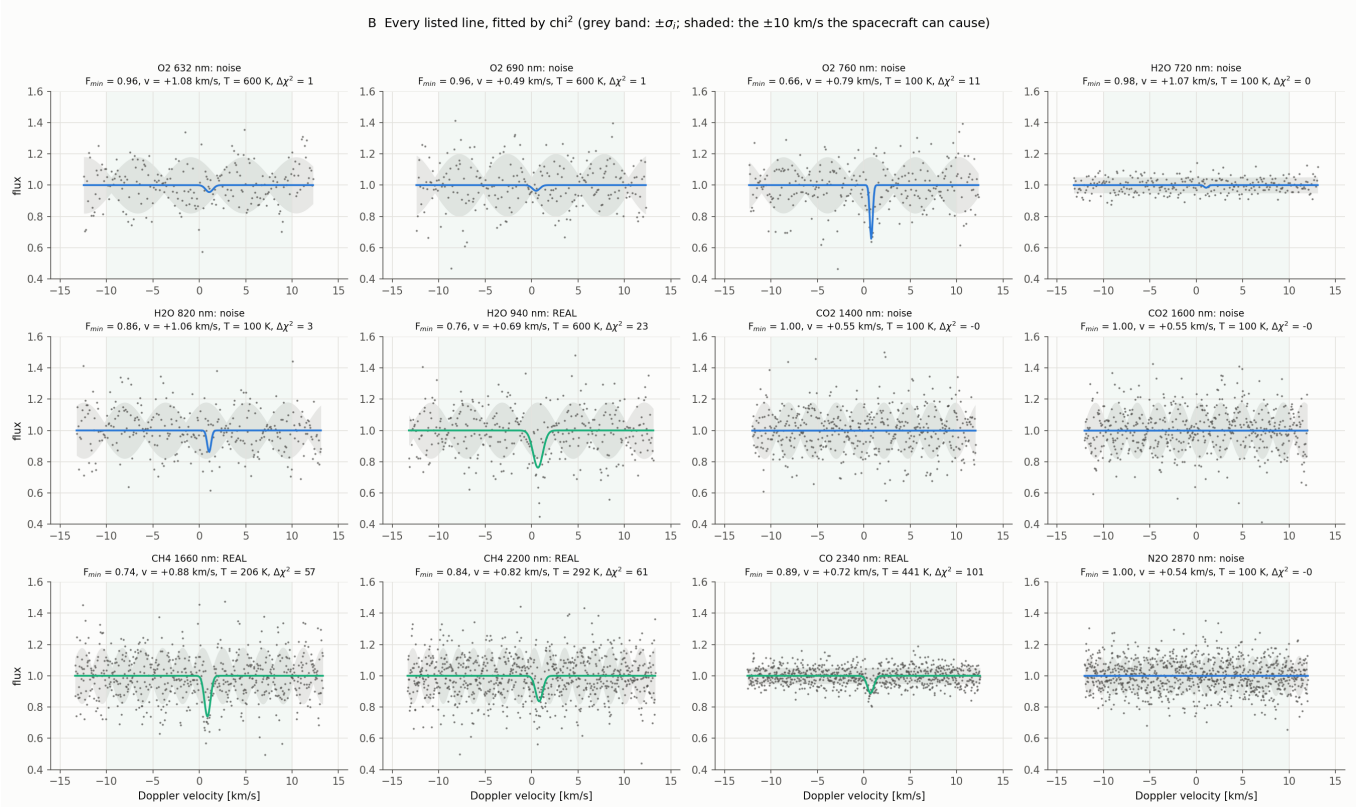


FIG. 6. Every listed line in its window on a Doppler-velocity axis, with the targeted fit. Green: detected.

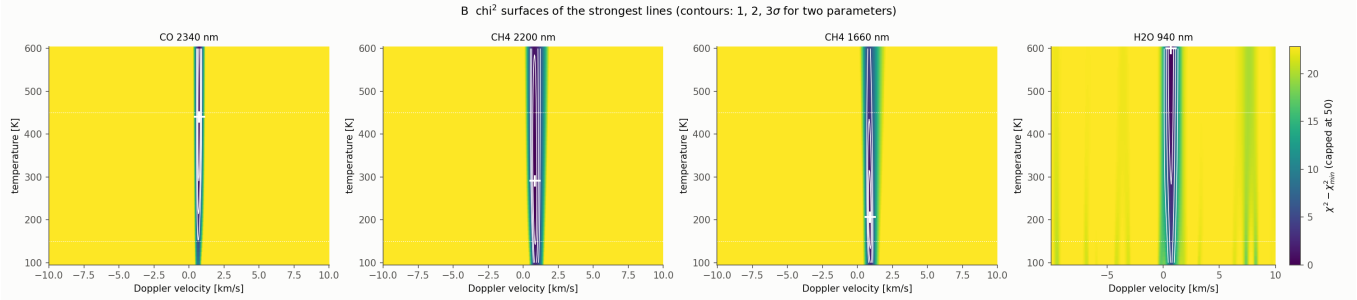


FIG. 7. χ^2 against Doppler velocity and temperature for the strongest lines (contours: 1, 2, 3σ).

- A safe orbit altitude.
- The atmosphere's scale height.
- The planet's spin.
- The absolute air density, decisive for the composition. The spectrum alone could not settle O_2: its lines fall where the noise is worst. A density measurement that needs $\mu \approx 23.63$ could.

The main uncertainties are as follows.

- **Assumed abundances.** Equal amounts of each gas are assumed, as instructed; the true abundances could differ.
- **Line temperatures.** These are poorly constrained.
- **The model's structure.** T_0 comes from a blackbody estimate, and the atmosphere is assumed to be adiabatic, then isothermal. The drag check tests all of these together in the upper atmosphere, not separately, and not near the surface where the landing happens.

Comparing with the API's hidden atmosphere afterwards confirms the model to within 2% below 300 km. The landing site (site 3) is confirmed for Part 7.

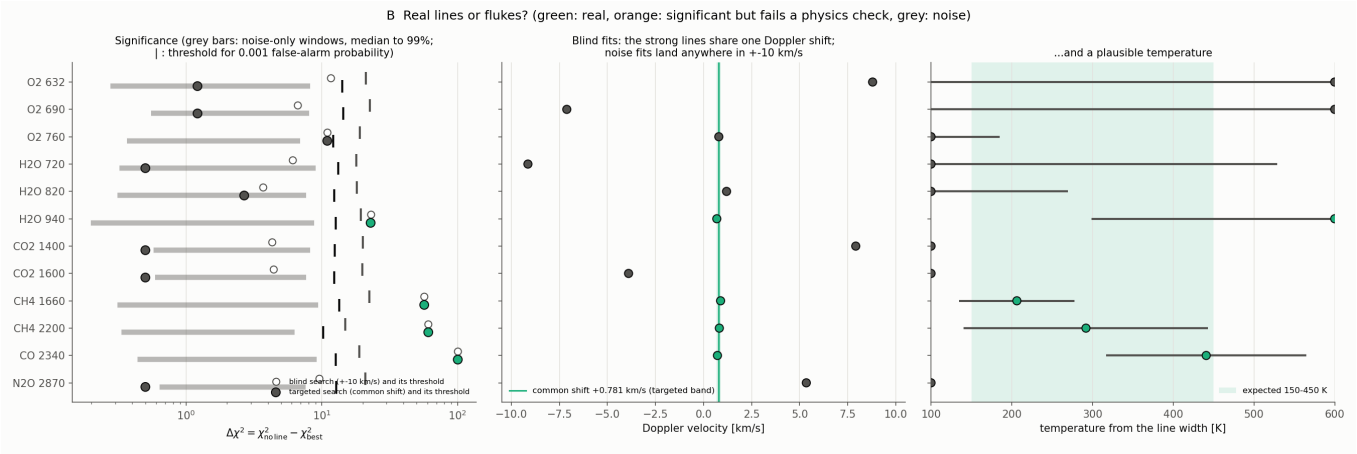


FIG. 8. Real lines or flukes. Significance in the blind and targeted searches against noise-only thresholds; blind Doppler velocities; temperatures.

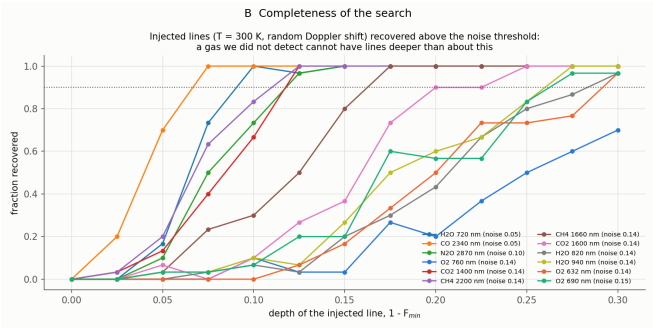


FIG. 9. Fraction of injected lines recovered against their depth, for every line position.

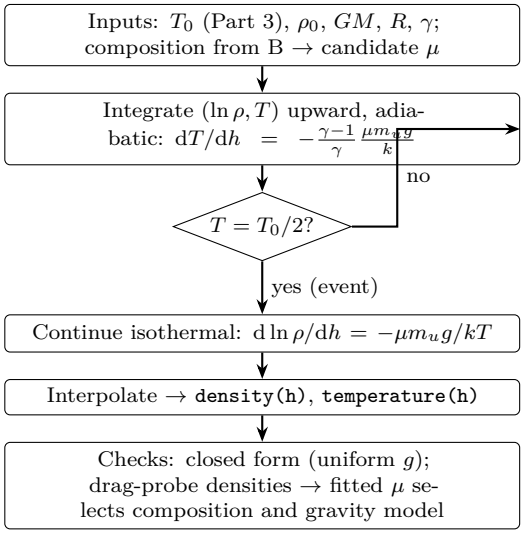


FIG. 10. Flowchart of the numerical atmosphere model (challenge C).

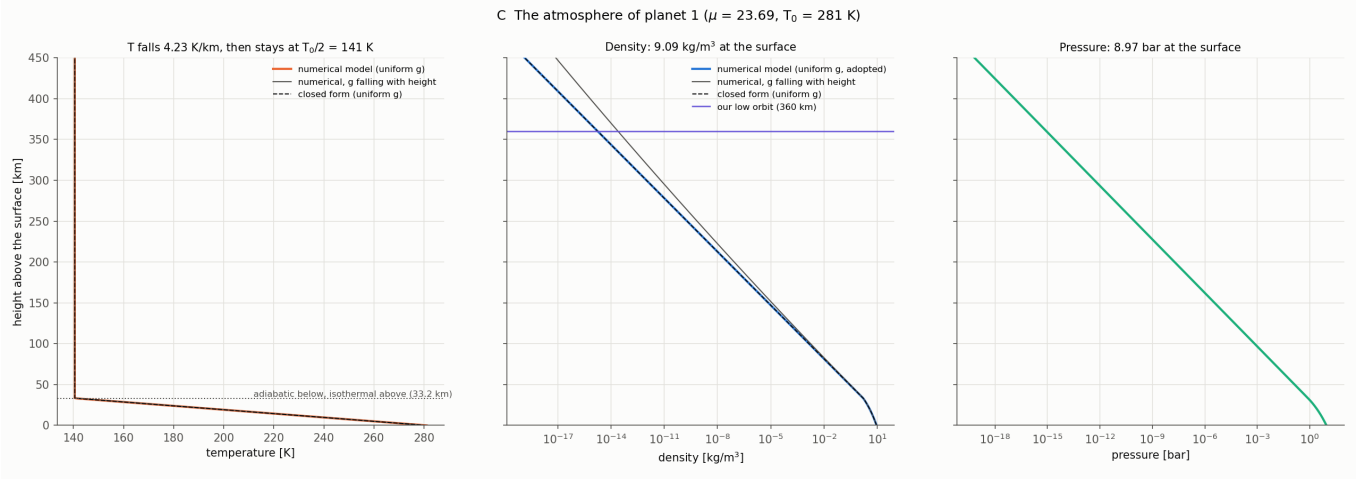


FIG. 11. Temperature, density and pressure against height. Numerical model, closed form, and the variant with gravity falling with height.

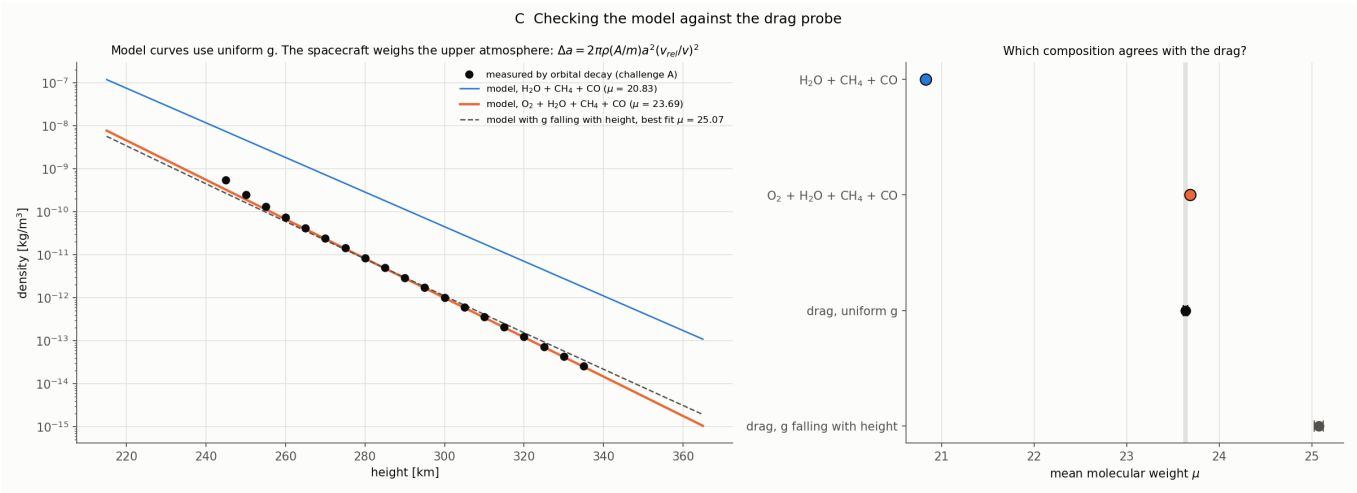


FIG. 12. Left: densities measured by orbital decay against the model for each allowed composition. Right: μ of each composition against the μ fitted to the drag data.

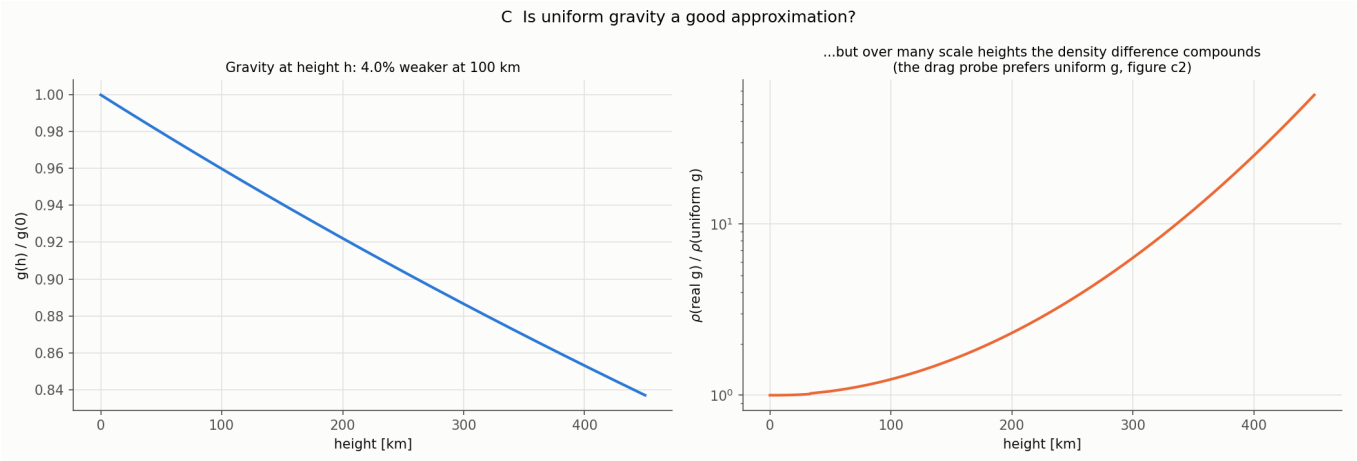


FIG. 13. How good is uniform gravity? $g(h)$, and the ratio of densities with real and uniform gravity.

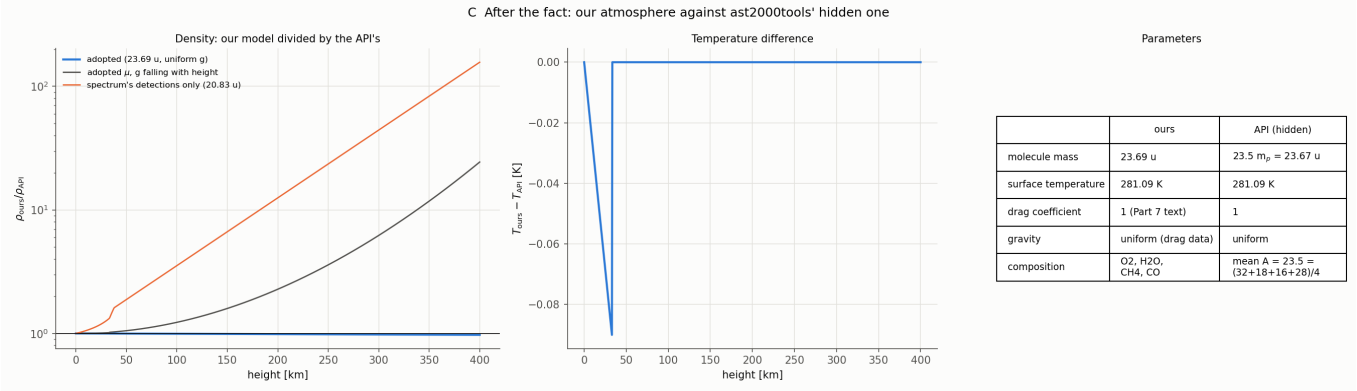


FIG. 14. After the fact: our density and temperature against the API's hidden atmosphere, and the parameters side by side.

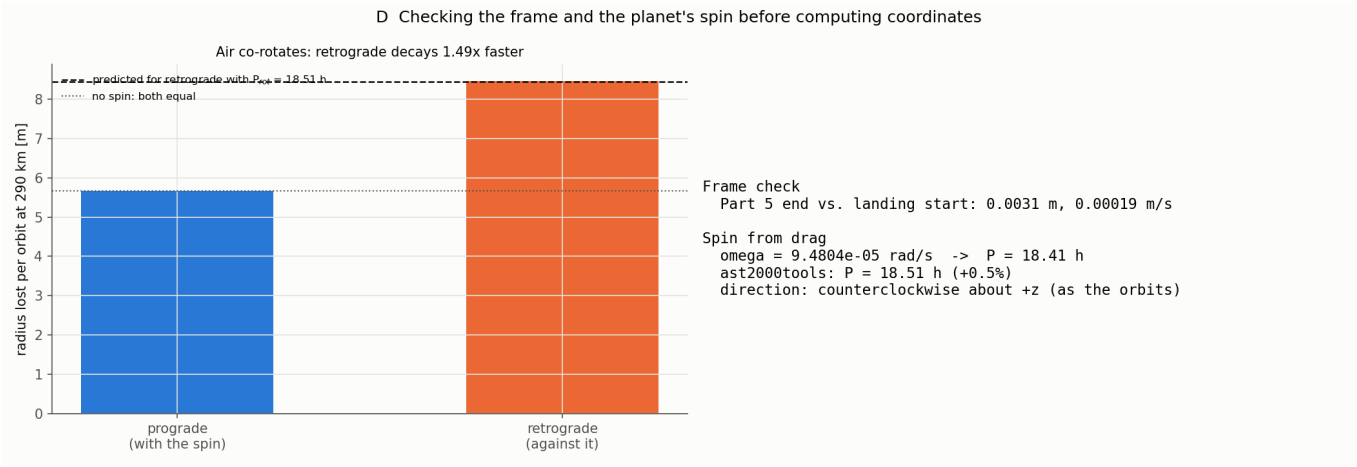


FIG. 15. Frame check and drag measurement of the planet's spin.

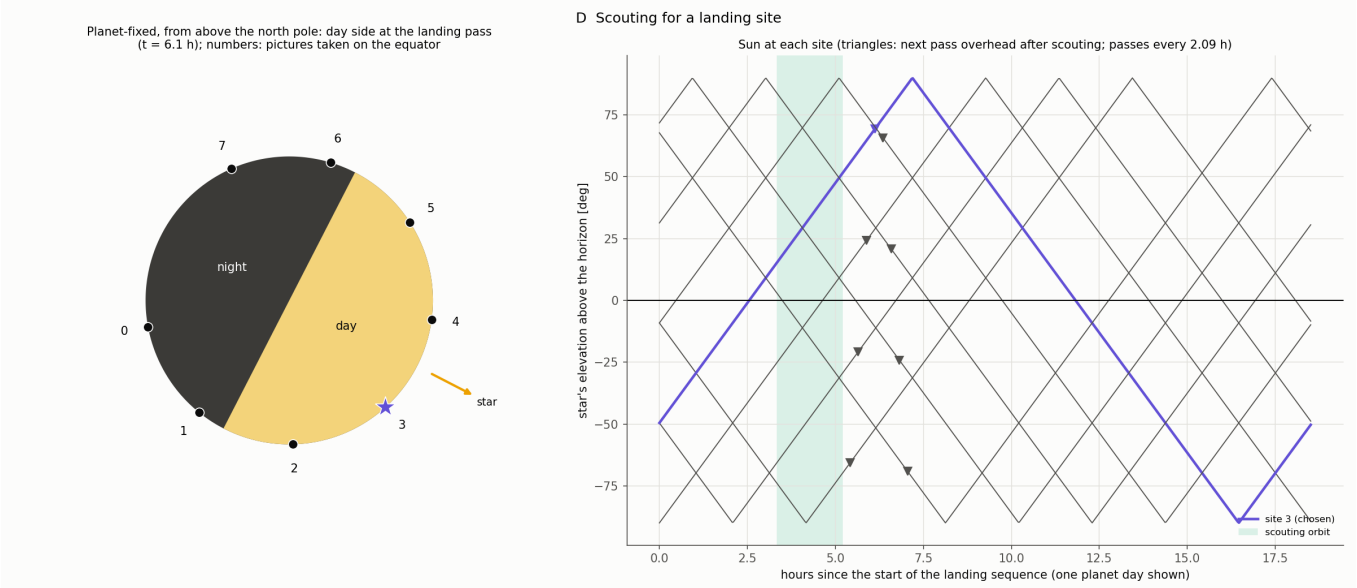


FIG. 16. Left: the planet seen from above its north pole in the planet-fixed frame, with day and night at the landing pass and the pictured sites. Right: star elevation at each site over one planet day, with the next pass overhead.

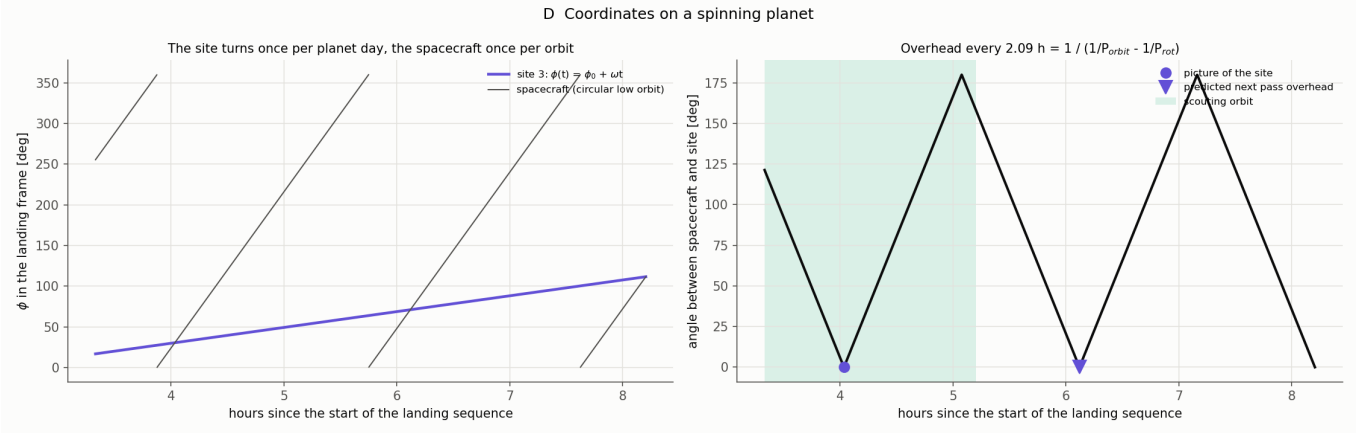


FIG. 17. The coordinate function at work: longitudes of the site and the spacecraft in the landing frame, and their separation with the predicted passes.

[1] F. K. Hansen and L. F. Engedal, ast2000tools: Python package for the AST2000 project, https://frodekh.github.io/ast2000tools/html/index.html (2023).

Appendix A: Mission log

Part 6 key numbers (seed 42716)	
Low orbit	360 km, period 112.5 min
Fuel left	2,235 kg
Drag scale height	9.26 km
Detected / marginal gases	H_2O, CH_4, CO / O_2
Common Doppler shift	+0.781 km/s
Adopted composition, μ	O_2 + H_2O + CH_4 + CO, 23.69
μ from drag	23.63±0.02
Surface P, ρ, T	8.97 bar, 9.09 kg/m^3, 281 K
Rotation period (drag)	18.41 h
Landing site	equator, $\phi = 312.0°$ (planet-fixed)